F70LA1 – Life Insurance Mathematics A
Chapter 7 – Policy Values

Read: Dickson, Hardy & Waters, Chapter 7.

Introduction

We now have different methods of calculating the premium required to cover benefits (and sometimes expenses) on a life insurance policy. This is an important calculation when running a life insurance company as premiums are often fixed, or guaranteed if you prefer, for the next twenty, say, years. However, during that time it is important for the company to consider all the contracts on its books and decide whether it has put enough money aside to meet the payments expected in the years to come. For this reason, actuaries calculate **Policy Values** and have to report them, amongst other things, to regulatory authorities.

Policy Values are also useful when recalculating the premium following the alteration of a policy at some point during its term.

Motivation

Example 1 *On 1 January 1990 a life aged 40 purchased a whole life insurance with sum insured £100,000 payable at the end of the year of death and annual premiums payable throughout the term. The premium basis is:*
Survival model: AM92 Ultimate
Interest: 6% per year
Expenses: 50% of the first premium and 10% of all subsequent premiums. £100 on payment of the sum insured.

(a) Calculate the annual premium.

(b) *At 1 January 2005, how much money does the life office* **need** *in respect of this policy?*

(c) *Calculate the corresponding premiums for a 15-year term insurance from age 40 and for a whole life insurance from age 55.*

Rationale

- When policies are issued to each member of a group of lives aged x the premiums are set so that they are expected to be sufficient to meet future benefits and expenses as they fall due.
- At a future time t, the premiums expected from that point forward *may not be* expected to be sufficient to meet future benefits and expenses as they fall due (for those policies that remain in force).
- The amount of money the life office needs at time t to provide for future benefits and expenses less future premiums is called the
  - **policy value**; or
  - **reserve**; or
  - **prospective reserve**; or
  - **prospective policy value**.
- The amount the company should have accumulated up to time t to provide for the **prospective policy value** is called the **asset share** and we will come to it later.

Policy values can be calculated **gross** or **net** of expenses, just as for premiums.

And ${}_{\mathbf{t}}\mathbf{V}$ is defined to be the **policy value** at time t.

Further Definitions

The **gross premium policy value** per policy in force at time t, is calculated as follows:

$$ {}_tV \equiv \mathrm{E}_t[PV_t(\text{future benefits} + \text{future expenses} - \text{future premiums})] $$

where E_t is the expected value **given that the policyholder is alive at time** t and PV_t is the present value **at time** t**.**

- The premium used in this calculation **is the actual** premium payable by the policyholder
- Care is needed with premiums, benefits and expenses payable **at** time t. Conventionally:
  - a premium or annuity payment payable at time t is a *future* payment; and
  - a death benefit payable *at* time t is a *past* payment.
- For with profits policies, the policy value should allow for **actual** bonuses already declared, and for future bonuses expected to be declared as per the policy value basis.

Note that the result obtained for the policy value depends on the basis used, a basis which may well be different from the one used to set the premium.

The **net premium policy value** per policy in force at time t, is calculated as follows:

$$ {}_tV = \mathrm{E}_t[PV_t(\text{future benefits} - \text{future premiums})] $$

where E_t is the expected value **given that the policyholder is alive at time** t and PV_t is the present value **at time** t**.**

- The EPVs are calculated on the net premium policy value basis, which would be specified.
- The premium used in this calculation is the **net premium** calculated on the **net premium policy value basis**, which may not be the premium payable by the policyholder.
- A net premium policy value is a special case of a gross premium policy value: ie one where there are no expenses and the premium is calculated on the net premium policy value basis.

Example 2 *Calculate the following policy values for the policy in Example 1 at 1 January 2007 just before the premium then due is paid and also just after the premium is paid.*

(a) The gross premium policy value using the premium basis.

(b) The net premium policy value using the basis:
Survival model: AM92 Ultimate
Interest 4% per year.

Example 3 *A life aged 35 purchases a* $20-$*year deferred life annuity of £1,000 per year to be paid continuously from age 55. Premiums are payable continuously for at most 20 years.*

(a) Calculate the annual amount of the premium and the gross premium policy value at durations 10, 20 and 30 years. The premium and gross premium policy value basis is:
Survival model: AM92 Select
Interest: 6% per year
Expenses: Initial expenses £200, renewal expenses of 5% of all premiums.

(b) Calculate the net premium policy value at durations 10, 20 and 30 years using AM92 Select, 4% interest.

Example 4

A life office sells a 10 year with profits endowment assurance policy to a life aged exactly 60.

The basic sum insured of £100,000 is payable at the end of the year of death or on survival to age 70.

Level premiums are payable annually in advance throughout the term of the policy.

Compound reversionary bonuses are declared at the start of each year, including the first.

Premium Basis:

Survival model: The force of mortality is given by $\mu_x = A + Bc^x$*, where:*

$$A = 0.0001; \quad B = 0.00035; \quad c = 1.075$$

Interest: 6% per year

Bonuses: 2% per year compound reversionary bonuses.

Calculate the annual premium.

The life office wants to calculate the gross premium policy value for this insurance just before the 6th premium is paid and the bonus for the coming year is declared.
The bonus additions made to date amount to £13000.
The gross premium policy value basis is:

Survival model: The force of mortality is given by $\mu_x = A + Bc^x$, *where:*

$$A = 0.0001; \quad B = 0.00035; \quad c = 1.075$$

interest: 8% per year

Future bonus additions: 2.5% per year compound

Calculate the gross premium policy value.

Recursion Formulae For Policy Values

We constructed recursion formulae for the expected present values of certain benefits which proved useful when combined with spreadsheets because they allowed us to get results for many different ages and terms quickly.

We can construct recursion formulae for policy values as well, and so see how policy values evolve over time.

Example 5
Consider a whole life insurance with sum insured £100,000, payable at the end of the year of death, issued to a life aged 40. Premiums are payable annually throughout life.

The basis for calculating the premium and gross premium policy values is:

Mortality: AM92 Select

Interest: 4% per year

Expenses: 5% of each gross premium and £100 when the sum insured is paid

(a) Calculate the gross premium, P.

(b) Calculate the gross premium policy value at duration 20, just before the premium then due is paid.

(c) Calculate the gross premium policy value at duration 21, just before the premium then due is paid.

(d) Show that for any non-negative integer duration, t,

$$({}_tV + 0.95P)(1+i) = 100\,100q_{[40]+t} + p_{[40]+t\ t+1}V$$

Recurrence Formula for Policies with Discrete (annual) Cashflows

Start by considering a policy issued to a (x) where cash flows – premiums, expenses and claims – can only occur at the start or end of a year.

Suppose this policy has been in force for t years, where t is a non-negative integer.

Consider the year $(t+1)$ and let:

P_t denote the premium payable at time t,

e_t denote the premium-related expense payable at time t,

S_{t+1} denote the sum insured payable at time $t+1$ if the policyholder dies in the year,

E_{t+1} denote the expense of paying the sum insured at time $t+1$,

${}_tV$ denote the policy value for a policy in force at time t,

${}_{t+1}V$ denote the policy value for a policy in force at time $t+1$,

q_{x+t} denote the probability that the policy holder, alive at time t, dies in the year, and

i_t denote the rate of interest assumed to be earned in the year.

The quantities $e_t,\ E_t,\ q_{x+t}$ and i_t are all as defined in the policy value basis.

Then:

$$({}_tV + P_t - e_t)(1+i_t) = q_{x+t}(S_{t+1} + E_{t+1}) + p_{x+t}\,{}_{t+1}V$$

Recurrence Formula for Policies with Continuous Cashflows

This recurrence formula is known as **Thiele's Differential Equation**. A version of it is included on page 37 of the Yellow Tables but it is a special case and so it is probably safer to derive it from first principles.

We start by considering a policy issued to (x) where cash flows occur continuously.

Suppose this policy has been in force for t years, where $t \in \mathbb{R}$.

Let:

- P_t denote the amount per year of premium payable at time t,
- e_t denote the amount per year of premium-related expense payable at time t,
- S_t denote the sum insured payable immediately at time t if the policyholder dies,
- E_t denote the expense of paying the sum insured at time t,
- ${}_tV$ denote the gross premium policy value for a policy in force at time t,
- μ_{x+t} denote the force of mortality for the policy holder at age $x+t$, and
- δ_t denote the force of interest assumed earned at time t.

The quantities $e_t,\ E_t,\ \mu_{x+t}$ and δ_t are all as defined in the policy value basis.

Consider an interval $0 \leq t < t+h$, where h is small, and consider the expected present value of cashflows from $t \to t+h$:

Recursive Calculation of Policy Values

Example 6 *Consider a 20–year endowment insurance issued to a life aged 30. The sum insured, £100 000, is payable at the end of the year of death and level annual premiums are payable throughout the term. The basis for calculating premiums and policy values is:*

Survival model: The force of mortality is given by $\mu_x = A + Bc^x$*, where:*

$$A = 0.0001; \quad B = 0.00035; \quad c = 1.075$$

Interest: $\delta = 0.04$ *per year*

Expenses: Nil

(a) Calculate the level annual premium, P*.*

(b) Explain why:

$$_{19}V = 100\,000\, e^{-0.04} - P$$

(c) Calculate $_tV$ *for* $t = 19, 18, \ldots, 1, 0$*.*

Euler's Method for Solving Differential Equations

In the case of continuously payable premiums and immediately payable benefits we need to solve Thiele's Differential Equation. There are a number of ways of doing this but the simplest is probably to apply Euler's Method.

Example 7 (Euler's Method) *Consider a 20–year endowment insurance issued to a life aged 30. The sum insured, £100 000, is payable immediately on death, or at the end of the term, and level premiums are payable continuously throughout the term.*

The basis for calculating premiums and policy values is:

Survival model: The force of mortality is given by $\mu_x = A + Bc^x$, *where:*

$$A = 0.0001; \quad B = 0.00035; \quad c = 1.075$$

Interest: $\delta = 0.04$ *per year*

Expenses: Nil

(a) Calculate the annual amount of the premium, P.

(b) Explain why:

$$\lim_{t \to 20^-} {}_tV = 100\,000$$

(c) Explain why for $0 \leq t < t + h < 20$*:*

$${}_tV \approx ({}_{t+h}V - h(P - 100\,000\,\mu_{30+t}))/(1 + h(0.04 + \mu_{30+t}))$$

(d) Calculate ${}_tV$ *for* $t = 19.95, 19.9, 19.85, \ldots, 0.05, 0$.

Comments

- The recursive calculation of policy values is very flexible: we can incorporate variable benefits, expenses, premiums, interest rates.
- Euler's method is crude.

  – Smaller values of h are likely to give better approximations, and this is the case if we extend example 7.

  – We could have also used the following, more accurate, approximation:

$$ {}_tV \approx ({}_{t+h}V(1 - h\,\mu_{30+t}) - h(P - 100\,000\,\mu_{30+t}))/(1 + 0.04\,h) $$

– There are better methods than Euler's to obtain numerical solutions to differential equations, *e.g.* a Runge–Kutta Method.

Asset Share

All of our policy value calculations so far have been **prospective policy values** which means we have been looking forward in time and thinking about what money we should hold to meet future benefits and expenses (less future premiums) as they fall due.

We now consider **asset shares**.

The **asset share** at duration t represents the cash the insurance company **actually has** at time t in respect of a policy still in force.

Compare this to a **prospective policy value** which is the amount of money an insurance company actually **needs**.

We define **asset share** as follows:

It is the accumulation to time t of past premiums received less past claims and past expenses actually paid, totalled over all identical policies, N say, which were simultaneously issued.

We have:

$$\textbf{Asset Share} \equiv \frac{\textbf{total accumulated funds}}{\textbf{number of survivors}}$$

Note that this formula has no N dependence.

If the following conditions are true:

- the premium is calculated using the equivalence principle;
- the premium basis is **the same as** the policy value basis;
- actual experience (ie that of survival functions, interest rates, expenses and bonus declarations etc.) is exactly as assumed in the premium basis and policy value basis;

then:

$$\textbf{Asset Share} = \textbf{Prospective Policy Value}$$

This identity is a very special case. You cannot assume it holds unless the above conditions are satisfied.

Example 8

A life office sells a 10 year endowment policy to a life aged exactly 60.

The sum insured of £100,000 is payable at the end of the year of death or on survival to age 70.

Level premiums are payable annually in advance throughout the term of the policy.

Premium and policy value basis:

Survival model: The force of mortality is given by $\mu_x = A + Bc^x$*, where:*

$$A = 0.0001; \quad B = 0.00035; \quad c = 1.075$$

Interest: 6% per year

Expenses: 10% of each premium

(a) Calculate the annual premium.

(b) Calculate the gross premium policy value at the start of the 4-th year.

(c) Calculate the asset share at the start of the 4-th year assuming that in the first 3 years:

Interest has been 7.5% per year;

1000 policies were sold. There were 15 deaths in the first year, 12 in the second year and 17 in the third year;

Expenses were 15% of the premiums in the first three years.

(d) Show that the asset share at the start of the 4-th year can be written:

$$((P-E)\ddot{a}^{*}_{60:\overline{3}|} - 10^{5}A^{*1}_{60:\overline{3}|})/A^{*\ 1}_{60:\overline{3}|}$$

assuming that in the first 3 years:

Interest has been i^* per year

The rate of mortality (for similar policies) has been according to some given survival model

Expenses per policy have been E in the first 3 years.

where * denotes calculation at rate i^* using the survival model actually experienced.

(e) Show that **if** *the experience has been as in the premium basis ($e^* = 0.1P$; $i^* = 0.06$ and the experienced survival model $\equiv$ the model in the premium basis) and* **if** *the policy value basis $\equiv$ the premium basis, then the* **asset share = policy value.**

Annual Profit

Here we think about the profit (or loss) that emerges from a policy each year.

Start by considering a large number, N, of identical life policies as they evolve from year t to year $t+1$, where t is an integer.

Define the following:

- The sum assured is S and it is payable at the end of the year of death.
- The annual premium is P.
- The interest rate is i per year.
- Expenses are e at time t.
- The survival probability is p_{x+t}.

Then, the recursive formula, from earlier in this chapter, multiplies up N times. We obtain:

$$N({}_tV \ + \ P \ - \ e)(1+i) \ = \ N(Sq_{x+t} \ + \ p_{x+t} \ {}_{t+1}V)$$

If our assumptions in respect of interest rate, expenses, survivors and deaths etc turn out to be correct then the

policy value + premiums − expenses

accumulated to time $t+1$ will be exactly what is required to pay

benefits at $t+1$

and set up the

policy values needed at $t+1$

for the following year.

However, if the assumptions turn out not to be correct (which is almost always the case) then the insurance company may have too much or too little money to set up the required policy values after paying claims.

- Too much money means that the company has a **surplus** or **profit**.
- Too little money means that the company has a **deficit** or **loss**.

Sources of Profit

Generally, for the above contract type, annual profit > 0 if:

- Interest earned is higher than expected;
- Expenses are lower than expected; and/or
- Fewer policyholders die than expected.

However, the last bullet, in particular, one must be careful with, because it only applies for insurances.

Profit is increased on an annuity contract if

- **more** policyholders die than expected.

That is why, in some countries, you can buy an *impaired* life annuity. The annuity is cheaper for you if you can prove that you are in a sufficiently poor state of health.

Annual Profit from Mortality, Expenses and Interest

Insurance companies are often interested in where profit arises from, for instance did we get our mortality assumptions wrong, were expenses higher than expected or did we make more on investing our premiums than we expected.

We can make answers to these questions clearer by manipulating our recurrence formula for N policies from above. We have:

$$N({}_tV + P - e)(1 + i) = N(Sq_{x+t} \ + \ p_{x+t} \ {}_{t+1}V)$$

Suppose now that:

- Actual expenses are e^* per policy at the start of the year;
- Interest is actually earned at rate i^* per year in the year; and
- n_d policyholders die in the year.

Then the cash available at the end of the year is given by:

$$N({}_tV + P - e^*)(1 + i^*)$$

and the cash required at the end of the year is:

$$n_dS + (N - n_d)\,{}_{t+1}V$$

Hence the profit in the year from mortality, expenses and interest together is:

$$N({}_tV + P - e^*)(1 + i^*) - n_dS - (N - n_d)\,{}_{t+1}V$$

Splitting out the Mortality Profit

Suppose $i^* = i$ and $e^* = e$. ie interest and expenses turned out as expected. It follows that the only source of profit is as a result of actual mortality being different to expected mortality.

We start with:

$$N({}_tV + P - e^*)(1 + i^*) - n_dS - (N - n_d)\,{}_{t+1}V$$

Now, applying $i^* = i$ and $e^* = e$ gives:

$$\begin{aligned} & N({}_tV + P - e)(1+i) - n_d S - (N - n_d)\,{}_{t+1}V \\ = \;& NSq_{x+t} + Np_{x+t}\,{}_{t+1}V - n_d S - (N - n_d)\,{}_{t+1}V \quad \text{by recurrence formula} \\ = \;& NSq_{x+t} + N(1 - q_{x+t})\,{}_{t+1}V - n_d S - (N - n_d)\,{}_{t+1}V \\ = \;& (Nq_{x+t} - n_d)(S - {}_{t+1}V) \quad \text{by cancelling } N\,{}_{t+1}V \text{ and collecting terms} \end{aligned}$$

This is a useful formula. We define the following:

- $S - {}_{t+1}V \equiv$ **Death Strain at Risk** (DSAR)
  You can think of it as the cost we must meet on death over and above the monies provided by the reserve at the end of the year.
- $Nq_{x+t}(S - {}_{t+1}V)$ is the **expected death strain at risk** (EDS)
- $n_d(S - {}_{t+1}V)$ is the **actual death strain at risk** (ADS)
- Therefore

$$\textbf{Mortality Profit} = \textbf{EDS} - \textbf{ADS}$$

Note that, for an annuity policy, S is zero and so the DSAR is negative.

Example 9 *On 1 January 1994 a life office issued a number of identical* 20 *year endowment insurances, with death benefits payable at the end of the year of death, to lives then aged 55. Each policy had a sum insured of £50 000 and annual premiums payable for at most 10 years.*

The premium and gross premium policy value basis is as follows:
Mortality: AM92 Select
Interest: 4% per year
Expenses: 10% of each gross premium

(a) Calculate the gross annual premium.

(b) On 1 January 2002 there were 200 of these policies still in force. In 2002: 5 policyholders died.
The actual expenses on 1 January 2002 were 12% of each gross premium.
The actual rate of interest earned in 2002 was $6\frac{1}{4}\%$.

Calculate the office's profit in 2002.

(c) On 1 January 2008 there were 180 of these policies still in force. In 2008: 4 policyholders died.
The actual rate of interest earned in 2008 was $5\frac{1}{2}\%$.

Calculate the office's profit in 2008.

Policy Alterations

Policyholders may request that their policy be altered. There are all sorts of things they could request and only some of them will be permitted within the policy conditions. Some examples are:

- Reduction of sum assured and so the premium payable
- Increase of sum assured and so the premium payable
- Increase in term
- Reduction in term
- Cessation of premium payment. This means that no more premiums will be paid. We call such a contract **paid up**. Altering a contract in this way will likely lead to a reduction in the sum assured. (The new sum assured is called a **paid–up sum assured**
- Changing the contract type. eg from a Whole Life Assurance to an Endowment Assurance
- Changing a policy from With Profit to Without Profit
- Cancelling (also known as lapsing or surrendering) a policy and accepting a sum of money now (known as the **surrender value**

And there are many others.

Note: Alterations are always at the request of the policyholder.

Basic Method for Calculating Altered Premiums/Benefits

Generally, altering a contract has consequences.

The sum assured or premium should change from the date of the alteration onwards.

We apply:

$$ {}_tV = {}_tV^* + C $$

where:

- ${}_tV$ is the policy value **before** making the alteration.
- ${}_tV^*$ is the policy value **after** making the alteration.
- C is the cost to the insurer of making the alteration.

The equation is then solved for the new P or S as required.

Note that a basis for the calculation needs to be specified.

If the basis is realistic then the insurer is choosing to neither make a profit nor a loss from the alteration.

Note: You should use this method, unless told otherwise.

Example 10 *A 20-year annual premium endowment insurance is issued to a life aged 40. Annual premiums of £500 are payable throughout the term and the sum insured of £15,000 is payable at the end of the year of death or at maturity.*
Immediately before the 10th premium is due the policyholder requests that no more premiums are paid.

(a) Calculate the surrender value payable immediately.

(b) Calculate the paid–up sum insured.

The basis for calculating revised benefits is:
Survival model: AM92 ultimate
Interest: 6% per year
Expenses: 10% of each premium payable and £100 for making the alteration.